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The perception and production of language-specific mid-vowel contrasts: Shifting the focus to the bilingual individual in early language input conditions

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Abstract

Aims and Objectives/Purpose/Research Questions: This study investigates the perception and production of the Catalan mid-vowel contrasts by 60 early Spanish-Catalan bilinguals addressing the following questions: (1) what are the effects of language dominance in the production and perception of the Catalan midvowels? (2) Does the perceptual deficit attributed in Barcelona remain in Majorca? (3) Do these bilinguals maintain the mid-vowel contrasts in production? And (4) what is the relationship between the perception and production of the mid-vowel contrasts within each bilingual individual?

Design/Methodology/Approach: Participants completed categorical AXB discrimination and picture-naming tasks to examine their perception and production of the Catalan mid-vowel contrasts.

Data and Analysis: The perception data consisted of 8,640 responses and the production dataset comprised 9,585 acoustic measurements that were submitted to mixed-model ANOVAs. Individual variation was explored by calculating the Euclidean distance between the mid-vowel categories for each speaker and exploring correlations with their accuracy rates in the perception task.

Findings/Conclusions: The results indicate that mid-vowels are more susceptible to discrimination difficulties than other vowel contrasts. Even though these bilinguals overall maintain robust midvowel contrasts in their productions, a closer examination reveals that the degree of language dominance affects the acoustic distance maintained between the mid-vowel targets. Individuals that produced the mid-vowels with smaller Euclidean distances were more likely to have a higher error rate in the perception task than bilinguals who produced a more robust contrast.

Originality: This study examines both mid-vowel contrasts in a variety of Catalan that has been reported to maintain robust mid-vowel contrasts in comparison to the acquisition deficit described in Barcelona, and explores the relationship between the production and perception abilities of each bilingual individual along a continuum of language dominance.

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Significance/Implications: This study contributes with new data to the study of language dominance and the relationship between the production and perception abilities of early and highly proficient bilinguals.

Keywords
Vowel perception/production, Language dominance, Mid-vowel contrasts, Individual variation, Early bilinguals

Introduction

Learning the sounds of a non-native language in a native-like manner is challenging, and it is typical for speakers of different languages to exhibit language-specific perception and production patterns reflecting their attunement to the phonology of their dominant language. Even after many years of experience, a "foreign" or L1-influenced accent is almost an expected outcome in the production (Flege, Munro, & MacKay, 1995; Piske, MacKay, & Flege, 2001) and perception (Jenkins, Strange, & Polka, 1995; Polka, 1991; Sheldon & Strange, 1982) of the L2.

Early bilinguals generally outperform late learners in different aspects of L2 pronunciation, as late learners typically face more difficulties in producing and perceiving native-like targets and are prone to exhibiting accent transfer from their L1 (DeKeyser, 2000; Flege, 1991, 1999, 2003, 2007; Lenneberg, 1967; Long, 1990; Oyama, 1976; Patkowski, 1990; Scovel, 1988; Yeni-Komshian, Flege, & Liu, 2000). Even though age of acquisition is a strong predictor for native-like attainment, it has been shown to not necessarily guarantee native-like proficiency (Flege & MacKay, 2004). A recurring example of phonological deficits despite early acquisition is of Spanish–Catalan bilinguals in Barcelona, who have been reported to have difficulties in acquiring the Catalan-specific mid-vowel contrasts: the four mid-vowels (/e/, /ɛ/, /o/, /ɔ/) are described as merging into two Spanish-like (/e/, /o/) mid-vowel categories, suggesting that early and extensive exposure to Catalan may not be sufficient to attain native-like phonetic abilities in the language.

The loss of the Catalan mid-vowel contrasts has been attributed to the extensive contact between Catalan and Spanish (Cortés, Lleó, & Benet, 2009; Lleó, Cortés, & Benet, 2008), and has been observed especially among younger generations in Barcelona, where /ɛ/ words are being pronounced with /e/ (Recasens, 1993, p. 86). In addition to their production patterns, difficulties in perceiving the Catalan /e/–/ɛ/ contrast have also been reported in a variety of behavioral tasks such as discrimination and typicality judgments (Bosch, Costa, & Sebastián-Gallés, 2000; Pallier, Bosch, & Sebastián-Gallés, 1997), gating tasks (Sebastián-Gallés & Soto-Faraco, 1999) and lexical decision tasks (Sebastián-Gallés & Díaz, 2012; Sebastián-Gallés, Echeverría, & Bosch, 2005). These studies show that the perception and production of these vowel contrasts remains difficult in spite of early and extensive exposure and everyday use of Catalan. This is a remarkable finding if we consider that these bilinguals have had many years of experience in their L2, they have been exposed to their L2 during the very first years of their lives, they have attained a high proficiency in the language, and they use both Spanish and Catalan on a daily basis. The question that remains is why the Spanish–Catalan bilinguals in Barcelona have difficulties acquiring these Catalan-specific phonemic categories.

Some studies suggest that only those speakers who have been exclusively exposed to (unaccented) Catalan since birth possess two phonemic categories (Pallier et al., 1997; Ramón-Casas, Swingley, & Sebastián-Gallés, 2009; Sebastián-Gallés & Soto-Faraco, 1999). That is, highly proficient bilinguals who were exposed to Catalan early in life, but were brought up in Spanish-speaking households, perceive the Catalan mid-vowels as a single Spanish-like vowel category,

resulting in functional "deafness" to the contrast. An alternative explanation that has been proposed is the lack of robustness of the Catalan mid-vowels rather than the loss of plasticity (Mora, Keidel, & Flege, 2011; Mora & Nadeu, 2012). This position argues that the Catalan mid-vowel contrasts are unstable because of several language-internal factors: 1. The mid-vowels vary across and even within dialects of Catalan. For instance, the word "to drink" in Eastern Catalan is pronounced /bɛwrə/ while the same word is /bewrə/ in western Catalan. 2. The implementation of mid-vowels may also vary across lexical items in a single dialect, for instance in the pronunciation of the word *res* "nothing" as /res/ or /rɛs/. This variation is observed within Barcelona, where there is a tendency to neutralize the mid-vowel contrasts due to the influence of Spanish (Recasens, 1993: p. 86), and in other varieties of Eastern Catalan (e.g. *Gironí, Sitgetà, Rossellonès*) where the mid-vowels have been described as near-mergers (Recasens & Espinosa, 2009). Also, 3. These mid-vowels have been reported to have a low functional load, as there are relatively few minimal pairs involving the front mid-vowel contrast. Finally, other research adopts a typological explanation arguing that complex and marked phenomena in one language may tend to be permeated by the other language if this is simpler and less marked (Lleó & Rakow, 2005; Lleó et al., 2008). The notion of markedness is related to frequency, with unmarked entities being more frequent than marked entities. In the case of vowel systems, having two degrees of height within the mid-vowels is less frequent than having just one degree (Maddieson, 1984). In other words, Catalan mid-vowels are vulnerable to Spanish influence because the Catalan vowel system is more marked (i.e. typologically less preferred) than the Spanish one.

Rather than being due to a lack of mental plasticity (Pallier et al., 1997; Ramón-Casas et al., 2009; Sebastián-Gallés & Soto-Faraco, 1999), the lack of robustness of the Catalan vowel system (Mora, Keidel, & Flege, 2011; Mora & Nadeu, 2012) or the potentially linguistically "marked" vowel system of Catalan (Lleó & Rakow, 2005; Lleó et al., 2008), these earlier findings may in fact be an artifact of the variety of Catalan being acquired. The study of early bilinguals in Majorca provides the opportunity to considerably reduce confounding factors that could have affected previous results in Barcelona. Due to differences in the historical evolution of the vowel systems in the dialects of Catalan, Majorcan Catalan has a phonological system and lexical distribution distinct from the variety spoken in Barcelona, and this variation particularly affects the Catalan mid-vowel contrasts. The variety of Catalan spoken in Majorca has eight rather than seven vowel phonemes, with an additional vowel phoneme /ə/, which appears in both stressed and unstressed positions[1] (Prieto, 2004; Recasens, 1991). Previous dialectal comparisons of Catalan have noted that there are differences between the vowel systems in Barcelona and other Catalan-speaking regions, showing that the magnitude of the acoustic distance between these mid-vowels is larger for Majorcan Catalan than for other dialects of Catalan: Majorcan Catalan /e/ and /o/ are produced lower than in Barcelona, while /ɛ/ and /ɔ/ are more "open," that is, they are produced with a higher first formant (F1) and lower second formant (F2) (Carrera-Sabaté & Fernández-Planas, 2005; Recasens & Espinosa, 2006; 2009).

Previous acoustic studies in Majorca have indicated that these bilinguals may be resisting the simplification of the vowel system that has been reported in Barcelona (Amengual, 2011, 2013; Herrick, 2003, 2006, 2007; Recasens & Espinosa, 2006, 2009; Simonet, 2014); see also Simonet (2011) for a merged Catalan mid-back vowel in the productions of Spanish-dominants). Given that recent acoustic studies have indicated that the magnitude of the acoustic distance between /e/–/ɛ/ and /o/–/ɔ/ is larger for Majorcan Catalan than for other dialects of Catalan, it is assumed that the input to which Majorcan Spanish–Catalan bilinguals are exposed contains more robust Catalan mid-vowel contrasts than the input available to bilinguals in Barcelona. As a result, it is predicted that Spanish-dominant bilinguals in Majorca will not show the perceptual "deafness" that has been reported in Barcelona and are also expected to maintain a significant acoustic distance between the mid-vowel targets in their productions, similarly to their Catalan-dominant counterparts. While much is known

about the perception and production patterns of Spanish–Catalan bilinguals in Barcelona, more empirical research is needed in a bilingual community where the mid-vowels have a different distribution and where a robust contrast may be more available in the ambient input all bilinguals receive.

Even though there is an undeniable link between speech production and perception, for the most part they have been independently investigated (Fowler & Galantucci, 2005). If production and perception reflect the properties of a unitary articulatory event (Pisoni, 1995), a central question is whether speakers can pronounce segments accurately in the L2 that they are not able to perceive well or whether accuracy in perception is a prerequisite to native-like pronunciation. The present study examines the production and perception abilities of a relatively large sample of early Spanish–Catalan bilinguals in Majorca representing a wide range of language dominance, and investigates individual variation in this bilingual community. What are the effects of language dominance in the production and perception of the Catalan mid-vowels? Are the mid-vowel contrasts phonologically "weak" and less robust than other vowel contrasts? Do these bilinguals maintain the contrasts in production? Does the perceptual deficit attributed to Spanish-dominant bilinguals in Barcelona remain in Majorca? And what is the relation between the perception and production of the Catalan mid-vowel contrasts within each bilingual individual? Individual variation has received little attention in the contact literature (Bullock, 2009). Assuming that "language change presupposes diffusion from individuals or smaller groups to the speech community as a whole" (Sankoff, 2013, p. 514), it is clear that the analysis of individual performance may help to disentangle patterns of individual variation in a bilingual context where the vowel contrasts are inherently variable.

Method

Participants

Sixty Spanish–Catalan bilinguals (33 females and 27 males) born, raised, and educated in Majorca[2] were recruited to participate in the perception and production experiments. Their ages ranged from 18 to 36 (M=21.5, SD=3.79), and all participants reported having extensive exposure to both languages on a daily basis, using Catalan and/or Spanish in the household, being educated in Catalan, and not being native in any other language. All participants reported normal speech and hearing and normal or corrected to normal vision, and each received a stipend for participating in the study.

Participants completed the Bilingual Language Profile (BLP) questionnaire (Birdsong, Gertken, & Amengual, 2012). The BLP is an instrument for assessing language dominance through self-reports and it produces a continuous dominance score and a general bilingual profile taking into account multiple dimensions: age of acquisition of the L1 and L2; frequency and contexts of use; competence in different skills; and attitudes towards each language. All of these factors are organized in four modules, which received equal weighting in the global language score: language history; language use; language proficiency; and language attitudes.[3] The BLP was administered prior to beginning the production and perception experiments, and was provided in Spanish or Catalan, depending on participant preference. The classification of participants as Spanish-dominant or Catalan-dominant was determined by the responses to the questionnaire, which generated a global score for each of the languages (Spanish and Catalan), a language particular score for each module, and a global score of dominance. The point system was converted to a scale score with the Catalan score subtracted from the Spanish score. As shown in Figure 1, Participants with negative points were classified as Spanish-dominant (N=30), while participants with positive points were classified as Catalan-dominant (N=30).

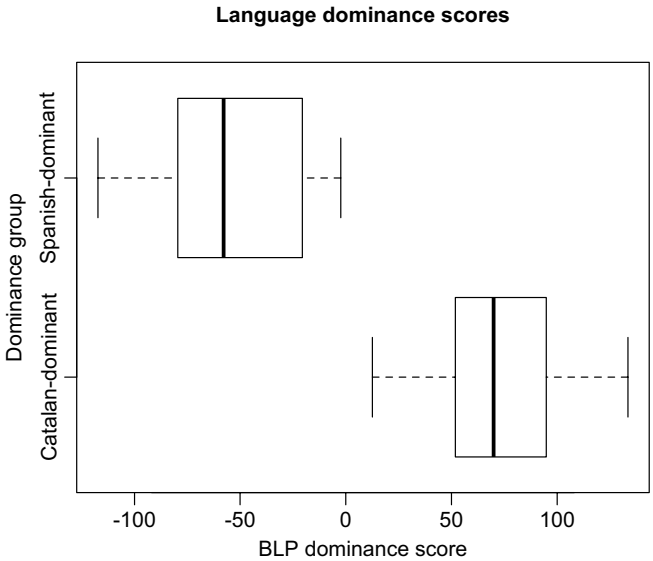

Figure 1. Language dominance as a function of group according to the BLP.
BLP: Bilingual Language Profile.

Table 1. Minimal pairs presented in the AXB discrimination task.

Vowel contrast	Minimal pair
/i/–/e/	fi "end" – fer "do"
/e/–/ɛ/	te "letter T" – te "tea"
/ɛ/–/a/	cel "sky" – sal "salt"
/u/–/o/	rus "Russian masc." – ros "blond masc."
/o/–/ɔ/	dóna "he/she gives" – dona "woman"
/ɔ/–/a/	por "fear" – pa "bread"

Note. The final /r/ in "fer" and "por" are silent.

Experiment 1: Perception of the mid-vowel contrasts

Materials and procedure. Participants completed a categorical AXB discrimination test to examine the robustness of the Catalan mid-vowel contrasts in comparison to other vowel contrasts in Catalan. In this procedure, A and B were naturally produced (i.e. non-synthesized) tokens of contrasting minimal pairs. Listeners were asked to select, for each trial, whether the middle item (X, or target) was the same as the first (A) or third (B) item. Because each of the three items presented in sequence was produced by a different speaker, the X was a physically different token than that of the matched A or B item. Therefore, listeners could not just make a simple acoustic identity judgment (Polka, 1991, 1992).

The stimuli consisted of 12 lexical items that formed six minimal pairs, including vowel contrasts across the front and back vowel region (Table 1). Even though unlike other Catalan varieties, Majorcan Catalan has minimal triplets with the mid-front vowels and /ə/ (e.g. /deu/ "God," /dɛu/ "ten," and /dəu/ "he/she owes"), the schwa (/ə/) was not considered in the tests because of the few minimal pairs in these contrasts and because the main goal of this experiment was to compare the perception of vowel

contrasts that existed both in Spanish and Catalan (e.g. /i/–/e/) to the Catalan-specific vowel contrasts (e.g. /e/–/ɛ/). A sequence of three words was created for each vowel contrast. In each triplet there were two words that were the same and one different, and the different word always appeared either first or last in the sequence. In other words, the stimulus that appeared in the middle of the sequence was always the same lexical item as the first or last item in the triplet. Therefore, there were a total of four possible combinations (four trials: AAB, BBA, BAA, and ABB), and because a different native Majorcan Catalan speaker produced each of the three words in a given triplet, there were a total of six possible speaker/voice combinations, for a total of 24 stimuli for each vowel contrast. The inter-stimulus interval was 1 second, and the inter-trial interval was 1.5 seconds.

Participants completed the AXB discrimination task in a quiet room using the stimulus presentation software *SuperLab Pro 4.5* (Cedrus Corporation, 2012) on a Mac computer. Participants listened to the three stimuli in sequence on a set of headphones under comfortable hearing conditions in front of a computer display, and were asked to decide whether the second stimulus (X) was more similar to the first (A) or to the last (B) by pressing the left or the right button on a USB Response Pad (RB-730). The experimental data consisted of 6 voice combinations × 4 trials × 6 vowel contrasts = 144 responses × 60 participants = 8,640.

Results. The data were analyzed under a mixed-model ANOVA with language dominance (Spanish-dominant, Catalan-dominant) as the between-subjects factor, contrast (/i/–/e/, /e/–/ɛ/, /ɛ/–/a/, /u/–/o/, /o/–/ɔ/, /ɔ/–/a/) as the within-subjects factor, and participant as the random term. The dependent variable was the error rate (%). The analysis yielded a main effect of *language dominance* ($F(1,58) = 13.31$, $p < 0.001$) and *vowel contrast* ($F(5,290) = 16.55$, $p < 0.001$), and a significant interaction between *vowel contrast* and *language dominance* ($F(5,290) = 4.12$, $p < 0.01$). To further explore the interactions, Bonferroni-corrected pairwise comparisons were run for each vowel contrast. Significant differences in the error rates between Catalan-dominant and Spanish-dominant bilinguals were found for the /o/–/ɔ/ contrast (diff. = −8.34, $t(29) = -2.47$, $p < 0.001$), the /e/–/ɛ/ contrast (diff. = −5.97, $t(29) = -2.21$, $p < 0.05$), and the /ɛ/–/a/ contrast (diff. = −5, $t(29) = -3.42$, $p < 0.05$), but there was no significant difference in language dominance for the /e/–/i/ contrast (diff. = −0.28, $t(29) = -0.26$, $p > 0.1$), the /ɔ/–/a/ contrast (diff. = −1.4, $t(29) = -1.22$, $p > 0.1$), or the /u/–/o/ contrast (diff. = −0.13, $t(29) = -0.21$, $p > 0.1$). Figure 2 shows the error rate (%) for each mid-vowel contrast as a function of language dominance.

These results indicate that some vowel contrasts, in particular the mid-vowel contrasts /e/–/ɛ/ (Catalan-dominant 5%, Spanish-dominant 11%) and /o/–/ɔ/ (Catalan-dominant 4%, Spanish-dominant 12.3%), together with /ɛ/–/a/ (Catalan-dominant 6.2%, Spanish-dominant 11.2%) displayed a significantly higher error rate than the other vowel contrasts. Additionally, in these same vowel contrasts there were significant differences between the Catalan-dominant and Spanish-dominant listeners, with Spanish-dominant bilinguals displaying a higher error rate. This supports previous findings suggesting that the Catalan mid-vowels (/e/–/ɛ/ and /o/–/ɔ/), which occupy the same acoustic-perceptual space as Spanish /e/ and /o/, may be particularly difficult for Spanish-dominant bilinguals to perceive. Interestingly, the results also indicate some difficulties distinguishing between the /ɛ/–/a/ pair. These apparent difficulties and differences between both groups may be due to a lower mid-vowel realization of /ɛ/ in Majorcan Catalan resulting in a shorter acoustic distance between /ɛ/ and /a/ in comparison to other Catalan varieties.

Experiment 2: Production of the mid-vowel contrasts

Materials and procedure. The production of the target Catalan mid-vowels was elicited in a picture-naming task. Pictorial representations of lexical items were selected instead of the written form to

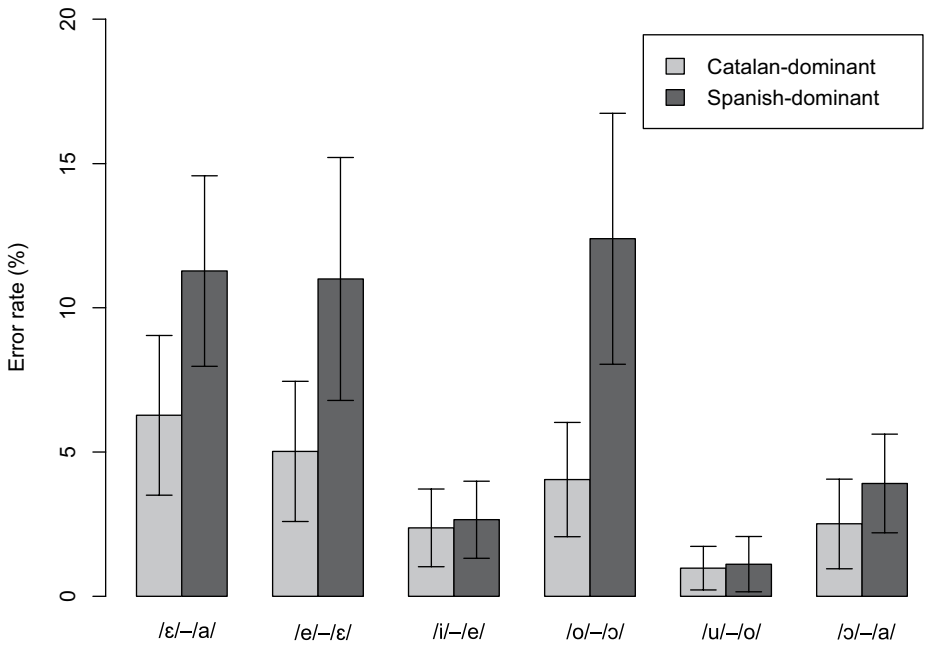

Figure 2. Error rate (%) for each vowel contrast by language dominance. Error bars enclose +/– one standard error.

Table 2. Sample stimuli from the picture-naming task.

Catalan	English	Target vowel	Picture
Bota	"boot"	/o/	
Flor	"flower"	/ɔ/	
Porc	"pig"	/ɔ/	
Poma	"apple"	/o/	

avoid orthographic effects in a reading-aloud task. The stimuli consisted of 40 experimental items that prompted the Catalan mid-vowels in stressed position. A sample of experimental items used in the picture-naming task is presented in Table 2.

The production task was conducted individually in a quiet room with participants comfortably seated in front of a computer display. Participants named pictures representing non-ambiguous objects (Snodgrass & Vanderwart, 1980) which appeared on a computer screen and their speech was recorded for subsequent acoustic analysis. The entire set of pictures was presented in randomized order and each picture appeared on a computer screen for five seconds together with the

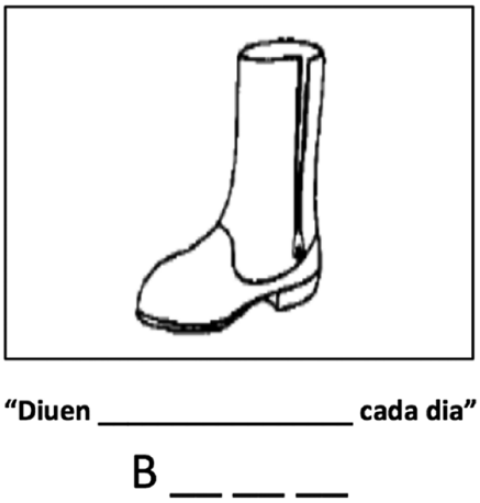

Figure 3. Visual stimuli for the picture-naming task.

first letter of the target word (Figure 3). Participants were asked to name the experimental word by embedding the target item in a carrier phrase, e.g. "Diuen *targetword* cada dia" – "(They) say *targetword* every day."

Because each picture appeared four times (once in each randomized block), each participant produced 160 experimental tokens for a total of 9,600 tokens. Fifteen tokens were excluded due to recording errors, or mispronunciations; as a result the dataset comprised a total of 9,585 measurements. The speech samples were recorded using a head-mounted microphone (Shure SM10A) and a solid-state digital recorder (Marantz PMD660), digitized (44kHz, 16 bit quantization), and computer-edited for subsequent acoustic analysis.

Acoustic analysis. Vowels were segmented in *Praat* (Boersma & Weenink, 2014) using synchronized waveform and spectrographic displays. *Praat* scripts were run to parse each participant's recording into individual files for each target item, and text grids were created by manually marking the vowel boundaries in each token. The boundaries of each vowel were determined by examining the waveform, spectrogram, and the intensity curve. Formant trajectories, especially the trajectory of the second formant ($F2$), as well as intensity displays, were taken as indicators of vowel onsets and offsets. The onset of the vowel was marked as the beginning of the first voiced cycle where $F2$ was visible and/or the intensity was similar to that of the vowel's midpoint (for voiceless obstruents), after the release (for voiced stops), the beginning of the first cycle in which $F2$ was visible and darkened (for fricatives), and at the beginning of the increase in intensity (for nasals and laterals). The end of the vowel was marked by the disappearance of $F2$, on the last pitch period (before stops and voiceless fricatives), the beginning of the decline in intensity (before nasals and laterals), and the onset of frication and decrease in intensity (before voiced fricatives). When the neighboring segment was an approximant, which preceded a vowel, the onset of the vowel was identified at the end of the transitional period between the approximant and vowel, where the vowel formants appeared as regions of stability in a steady-state pattern.

Vowel formant measurements were automatically extracted at the center of the steady-state period of the vowel. Formant tracks were calculated with the Burg algorithm as built into the *Praat* program. The effective window length for the calculation was set at 25ms, and was maintained

Table 3. Results from the mixed-model ANOVAs and pairwise comparisons on vowel height (*F*1) and frontness (*F*2) values for female and male Catalan-dominants (CAT) and Spanish-dominants (SPN).

FEMALES	/e/–/ɛ/	*F1*	– *vowel* ($F(1,32) = 270.17$, $p < 0.001$) – *group* ($F(1,32) = 0.04$, n.s.) – *vowel*group* ($F(1,32) = 6.56$, $p < 0.05$)	– *CAT*: diff.=−1.48, $t(25) = -25.93$, $p < 0.001$ – *SPN*: diff.=−1.08, $t(41) = -11.76$, $p < 0.001$
		F2	– *vowel* ($F(1,32) = 86.60$, $p < 0.001$) – *group* ($F(1,32) = 0.02$, n.s.) – *vowel*group* ($F(1,32) = 0.53$, n.s.)	– *CAT*: diff.=0.90, $t(25) = -21.39$, $p < 0.001$ – *SPN*: diff.=0.77, $t(41) = 7.78$, $p < 0.001$
	/o/–/ɔ/	*F1*	– *vowel* ($F(1,32) = 134.21$, $p < 0.001$) – *group* ($F(1,32) = 1.06$, n.s.) – *vowel*group* ($F(1,32) = 8.54$, $p < 0.01$)	– *CAT*: diff.=−1.35, $t(25) = -24.84$, $p < 0.001$ – *SPN*: diff.=−0.82, $t(41) = -8.38$, $p < 0.001$
		F2	– *vowel* ($F(1,32) = 198.93$, $p < 0.001$) – *group* ($F(1,32) = 0.10$, n.s.) – *vowel**group ($F(1,32) = 4.09$, n.s.)	– *CAT*: diff.=−0.86, $t(25) = -6.66$, $p < 0.001$ – *SPN*: diff.=−0.65, $t(41) = -6.38$, $p < 0.001$
MALES	/e/–/ɛ/	*F1*	– *vowel* ($F(1,24) = 295.59$, $p < 0.001$) – *group* ($F(1,24) = 0.35$, n.s.) – *vowel*group* ($F(1,24) = 3.86$, n.s.)	– *CAT*: diff.=−1.20, $t(33) = -20.83$, $p < 0.001$ – *SPN*: diff.=−0.93, $t(17) = 9.99$, $p < 0.001$
		F2	– *vowel* ($F(1,24) = 184.25$, $p < 0.001$) – *group* ($F(1,24) = 0.05$, n.s.) – *vowel*group* ($F(1,24) = 3.443$, n.s.)	– *CAT*: diff.=0.86, $t(33) = 15.61$, $p < 0.001$ – *SPN*: diff.=0.64, $t(17) = 9.96$, $p < 0.001$
	/o/–/ɔ/	*F1*	– *vowel* ($F(1,24) = 163.33$, $p < 0.001$) – *group* ($F(1,24) = 0.04$, n.s.) – *vowel*group* ($F(1,24) = 0.44$, n.s.)	– *CAT*: diff.=−0.86, $t(33) = -19.61$, $p < 0.001$ – *SPN*: diff.=−0.77, $t(17) = -7.25$, $p < 0.001$
		F2	– *vowel* ($F(1,24) = 161.14$, $p < 0.001$) – *group* ($F(1,24) = 0.76$, n.s.) – *vowel*group* ($F(1,24) = 0.73$, n.s.)	– *CAT*: diff.=2.13, $t(33) = 15.44$, $p < 0.001$ – *SPN*: diff.=−0.65, $t(17) = -4.40$, $p < 0.001$

*F*1: formant 1; *F*2: formant 2

across tokens and speakers. The maximum number of formants to be located by the formant tracker was always five, and the ceiling was set at 5.0 kHz for males and 5.5 kHz for females. These gender-specific formant ceilings reflect the different average vocal tract lengths of men versus women and were deemed appropriate after visual inspection of the sound files. Formant values were extracted in Hertz and were further converted to Bark (Traunmüller, 1990; Zwicker, 1961).

It has been acknowledged that the raw formant frequencies of different speakers are not directly comparable because of inter-speaker variation due to inherent physiological or anatomical differences (Clopper, 2009; Watt, Fabricius, & Kendall, 2010). In order to permit accurate cross-speaker comparisons of formant data, a variety of vowel formant normalization procedures have been proposed (Adank, Smits, & van Hout, 2004). Because there is still a lack of consensus over which available normalization algorithm is best to use and normalization techniques have been acknowledged to have drawbacks, the preferred option for this study was a more conservative one: investigating the formant values for males and females separately.

Results. Datasets of by-subject aggregates for females and males were created including the median *F*1 and *F*2 values over subjects as a condition of vowel (four values per participant).[4] The *F*1 and *F*2 mid-vowel data were submitted to mixed-model ANOVAs with group (Spanish-dominant, Catalan-dominant) as the between-subjects factor, vowel (/e/ and /ɛ/ for the front vowels; /o/ and /ɔ/ for the back vowels) as the within-subjects factor, and participant as the random term. Additionally, interactions were explored by analyzing the effects of vowel for each language dominance group separately in Bonferroni-adjusted pairwise comparisons. Table 3 presents the results of the ANOVAs and pairwise comparisons on the *F*1 and *F*2 values for the front

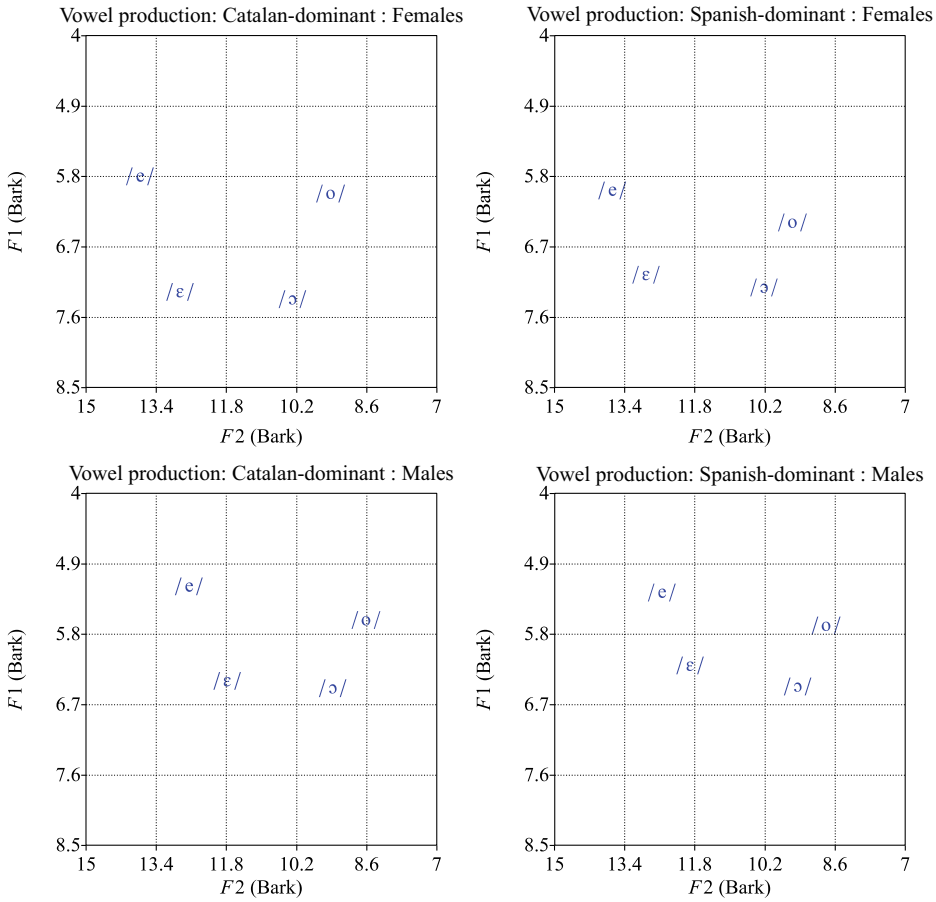

Figure 4. *F*1 × *F*2 vowel charts plotting the Catalan front and back mid-vowels produced by female and male Catalan-dominant and Spanish-dominant bilinguals.

and back mid-vowel data (females and males separately). Figure 4 displays four *F*1 × *F*2 vowel charts plotting the Catalan front and back mid-vowels produced by female and male Catalan-dominant and Spanish-dominant bilinguals.

Inspection of the vowel plots suggests that both Catalan-dominant and Spanish-dominant bilinguals implement a Catalan-specific /e/–/ɛ/ and /o/–/ɔ/ contrast, and results from the ANOVAs show that these Spanish–Catalan bilinguals in Majorca, who have been exposed to both languages from a very early age, produce robust mid-vowel contrasts in Catalan. Because the presentation of group averages may obscure distinct patterns of between-speaker variation, the next step was to examine the extent to which the mid-vowel contrasts are realized for each individual speaker.

In order to investigate the individual variation in the data, the Euclidean distance (in Bark), based on the median *F*1 and *F*2, between /e/–/ɛ/ and /o/–/ɔ/ was calculated for each individual participant. Euclidean distances were used to represent the two-dimensional distance between the F1 and F2 values for the two mid-vowels, as calculated in the following formula:

$$d\,(p, q)=\sqrt{(p1-q1)^2+(p2-q2)^2}$$

The Euclidean distance provides a quantitative measure to examine how robustly the contrast is maintained in the productions of each participant (see Appendix 1 for the formant values and Euclidean distances in Bark for each individual). Speakers that are merging the vowels and therefore losing the contrast between these Catalan mid-vowels have a smaller value than those who maintain a more robust contrast based on the distance between both mid-vowel targets in their productions. The Euclidean distance value was correlated with the participants' language dominance score as reported in the BLP (see Figure 5). The correlations between BLP score and Euclidean distance of the /e/–/ɛ/ contrast was not significant for the Catalan-dominant bilinguals (n=30, df=28, r=.32, $R2$=.10, n.s.); however, there was a highly significant positive correlation for the Spanish-dominant group (n=30, df=28, r=.55, $R2$=.30, $p < 0.01$). In addition, correlations between BLP score and Euclidean distance of the /o/–/ɔ/ contrast also yielded a non-significant correlation for the Catalan-dominant bilinguals (n=30, df=28, r=.21, $R2$=.04, n.s.), but a highly significant positive correlation for the Spanish-dominant group (n=30, df=28, r=.68, $R2$=.47, $p < 0.001$).

As Figure 5 shows, the Euclidean distances are in general smaller for Spanish-dominant bilinguals (negative BLP score) than for Catalan-dominant bilinguals (positive BLP score), although there is a certain amount of overlapping between the Spanish-dominants that are approximating the 0 score (balanced bilinguals) and the rest of the Catalan-dominant bilinguals. Notice that the data from the Spanish-dominant group are rather different from the Catalan-dominant group in that more individuals are merging their mid-vowel categories. For instance, nine Spanish-dominant participants (14, 18, 25, 30, 33, 36, 38, 47, and 54) produced their front mid-vowels with Euclidean distances smaller than one Bark, while there were not any in the Catalan-dominant group. Similarly, 11 Spanish-dominant participants (3, 5, 6, 14, 25, 30, 33, 36, 38, 47 and 54) and only three Catalan-dominants (11, 19 and 31) produced the back mid-vowels with Euclidean distances smaller than one Bark. Based on the information from their language background questionnaires it was found that those who did not produce robust mid-vowel contrasts had reported an exclusive use of Spanish at home, a more restricted and limited use of Catalan on a daily basis, and had also self-rated their Catalan pronunciation as being strongly influenced by Spanish. This supports previous findings that L2 use and amount of exposure to the language plays a crucial role in the production of vowel contrasts (Mora & Nadeu, 2012).

In order to gain more insight into the individual variation of the data, the relationship between the production and perception patterns of each bilingual individual were explored by correlating the Euclidean distance between the front and back mid-vowel categories with the percentage of correct responses to each mid-vowel contrast in the AXB discrimination task. Figure 6 plots the accuracy rate in the AXB discrimination task as a function of the Euclidean distance between the same speaker's /e/–/ɛ/ and /o/–/ɔ/ productions. Inspection of the plots shows that participants that overall had a lower accuracy rate in the perception task were also maintaining a smaller acoustic distance between the mid-vowels in their productions. These participants reported a more frequent daily use of Spanish over Catalan and their scores were identified as ranging between –41 and –117, therefore moderately to strongly Spanish-dominant in the BLP. The Euclidean distance between /e/–/ɛ/ and the accuracy rates in the AXB discrimination task provided a significant correlation (n=60, df=58, r=.52, $R2$=.27, $p < 0.001$). There was also a positive correlation between the /o/–/ɔ/ Euclidean distance and the accuracy rates in the AXB discrimination task (n=60, df=58, r=.44, $R2$=.19, $p < 0.001$). Even though this positive correlation was moderate, it reveals that participants who had a higher accuracy rate in the AXB discrimination task were also producing front mid-vowel categories with a larger acoustic distance.

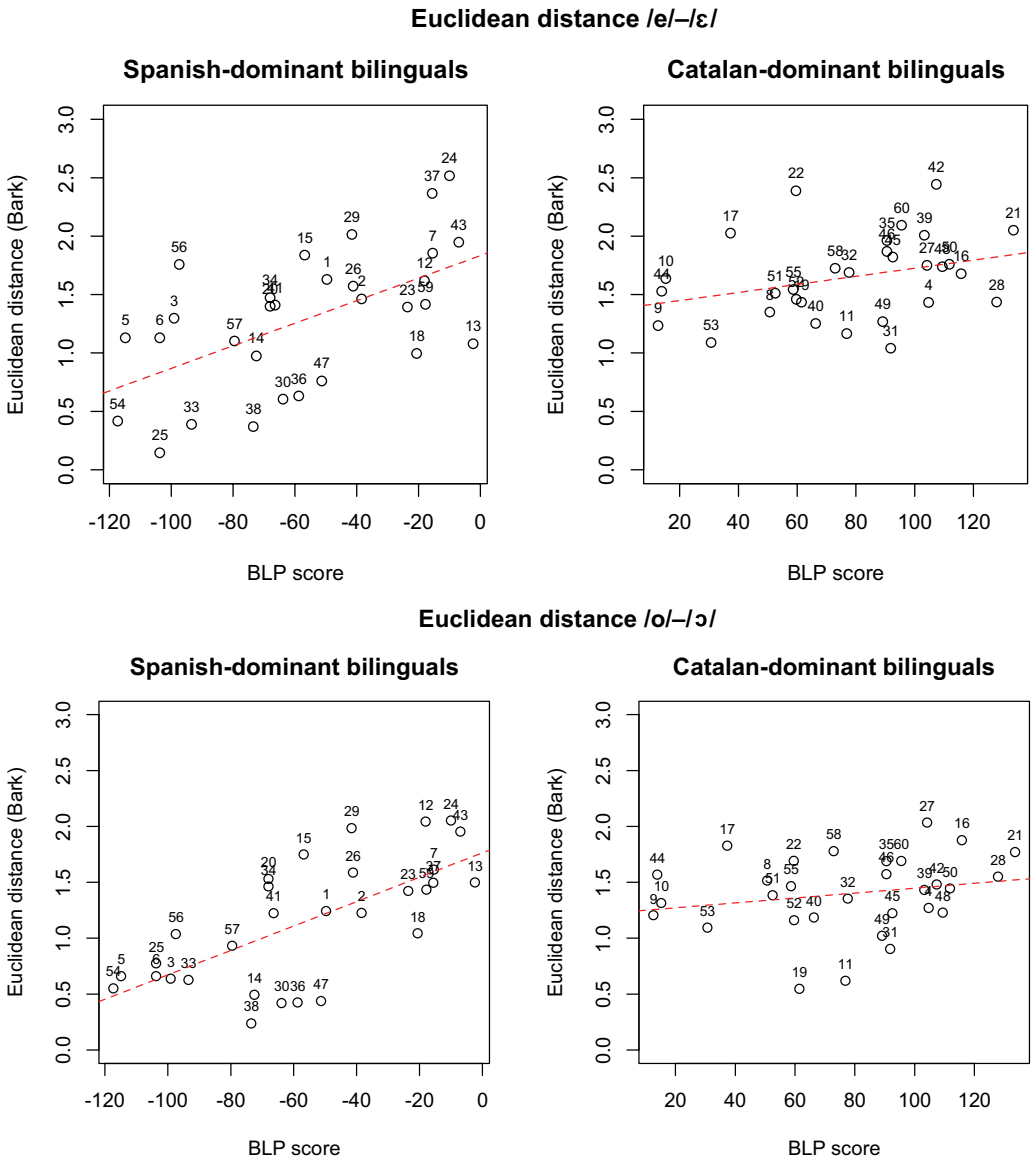

Figure 5. Euclidean distance between same speaker tokens of the vowels /e/–/ɛ/ and /o/–/ɔ/ plotted as a function of a speaker's BLP score. Fitted lines for the Spanish-dominants (left) and Catalan-dominants (right).
BLP: Bilingual Language Profile.

Discussion and conclusion

The present study examined the perception and production of the Catalan mid-vowel contrasts /e/–/ɛ/ and /o/–/ɔ/ by 60 early and highly proficient Spanish–Catalan bilinguals in Majorca. Although the Catalan /e/–/ɛ/ contrast has received a lot of attention in previous studies in Barcelona, a review of the literature revealed the lack of studies that examine individual variation in the

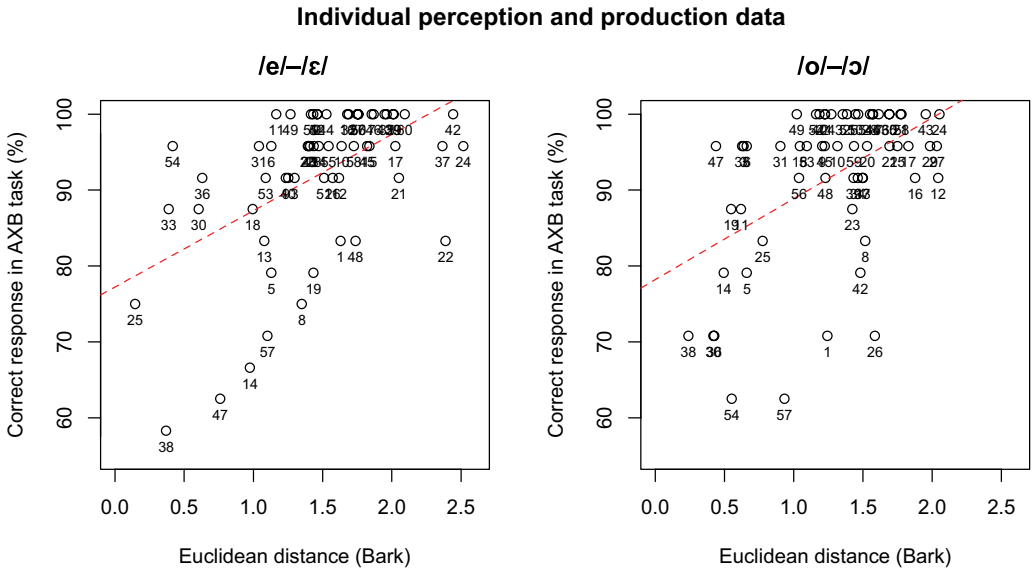

Figure 6. Accuracy rates in the AXB discrimination task plotted as a function of the Euclidean distance between the same speaker's /e/–/ɛ/ (left) and /o/–/ɔ/ (right).

perception and production of both mid-vowel contrasts in a variety of Catalan that has been reported to maintain a larger acoustic distance between the mid-vowel phonemes. The present study also goes beyond previous work in that it explores the interaction between the production and perception abilities of each bilingual individual along a continuum of language dominance.

The results from previous research indicate that the mid-vowels are more susceptible to discrimination difficulties, particularly by Spanish-dominants, than other vowel contrasts in Catalan. This is attributed to the fact that the Catalan mid-vowels are acoustically similar to Spanish /e/ and /o/. The AXB discrimination experiment was designed to test if Spanish-dominant bilinguals are able to perceive Catalan-specific contrasts as accurately as Catalan-dominants, and if these mid-vowels are harder to perceive than other vowel contrasts in Catalan. The results suggest that these bilinguals possess two independent phonetic categories in the mid-vowel space, and as a result perceive both Catalan-specific categories. Despite a relatively low error rate, the highest proportion of errors was found in the front and back mid-vowel contrasts and the /ɛ/–/a/ contrast. Because of a lower production of /ɛ/ in Majorcan Catalan, the errors with the /ɛ/–/a/ contrast are argued to be due to a reduced acoustic distance between /ɛ/ and /a/ in this variety of Catalan.

This study also investigated the acoustics of the Majorcan Catalan mid-vowels, and compared the productions of Catalan-dominant and Spanish-dominant bilinguals. The results from the picture-naming task revealed that both groups overall maintained robust mid-vowel contrasts in their Catalan productions. Although there was more variability in the spectral characteristics of the mid-vowels produced by the Spanish-dominant group (i.e. with a subset of the group maintaining a smaller Euclidean distance between the mid-vowel categories), most Spanish-dominants in this study produced Catalan mid-vowel contrasts that did not significantly differ from those produced by Catalan-dominants. A closer examination of individual differences indicated that strongly Spanish-dominant bilinguals who used and were exposed to more Spanish than Catalan maintained a smaller acoustic distance between the mid-vowels than moderately Spanish-dominant ones. This

suggests that the effect of Spanish dominance on the production of Catalan vowels is gradient and strongly affected by the individual's amount of use and exposure to Catalan and Spanish.

The present study also examined the relation between the production and perception abilities of each bilingual individual. The correlations between the production data and the results from the AXB discrimination task indicated that bilinguals who had a higher accuracy rate in the AXB discrimination task were producing the Catalan mid-vowels with a larger Euclidean distance. These results support previous findings suggesting that native-like production is limited by how accurately the sounds are perceived (Flege, 1993; Flege, Bohn, & Jang, 1997; Flege, MacKay, & Meador, 1999). It must be noted that the ability to draw strong conclusions from these correlations is hampered by ceiling effects, as the bilinguals in this study achieved a high accuracy rate in both the production and perception tasks and most of the variability in the data was centered on a reduced subset of the most strongly Spanish-dominant bilinguals. To address these potential limitations, further analyses exploring the performance of bilinguals who only partially succeed in the perception and/or production tasks will provide richer data to confirm the correlation between the perception and production of language-specific sounds in early bilinguals.

One of the main goals of the present study was to investigate if speakers of a dialect that has a phonological system distinct from the variety spoken in Barcelona have difficulties in acquiring Catalan-specific vowel contrasts. The present findings do not support previous accounts of "severe limitations in the acquisition of Catalan-specific categories under conditions of early and extensive exposure" (Sebastián-Gallés & Soto-Faraco, 1999, p. 120) but they do show that despite the acoustic similarity with the Spanish mid-vowel categories, these early Spanish–Catalan bilinguals in Majorca are capable of maintaining language-specific contrasts, even for acoustically similar phones (MacLeod & Stoel-Gammon, 2010).

As noted in Flege (2007), there is increasing evidence that the quantity and quality of L2 input received by an L2 learner is an important determinant of ultimate degree of attainment in an L2. In Barcelona, Spanish-like Catalan mid-vowels have been reported especially among younger generations, where there is a tendency to pronounce /ɛ/ words with /e/ (Recasens, 1993, p. 86). Spanish-dominant bilinguals in Barcelona who often fail to produce an effective contrast between the mid-vowel pairs are reportedly leading this change by providing accented input to younger generations (Mora & Nadeu, 2012). In line with this approach, several studies have identified the quality of the input as the main predictor of native-like production patterns in the acquisition of children and adults in Barcelona, demonstrating that the maintenance of the mid-vowel contrasts is closely related to the region, district, and social network of the speakers (Cortés et al., 2009; Lleó, Benet, & Cortés, 2007, 2009; Lleó et al., 2008). These studies show that it is in highly Spanish-influenced districts in Barcelona where bilinguals are merging the Catalan mid-vowel categories in their productions due to being exposed to more variable input. As a result, in a bilingual language contact setting such as Barcelona, Spanish-dominant speakers may receive highly variable and inconsistent Catalan input (i.e. Spanish-accented Catalan), which in terms of the Catalan mid-vowels leads to difficulties in the acquisition of the contrast (Bosch & Ramón-Casas, 2011). This is not just an example of a contrast that exists in one language, Catalan, and not in the other, Spanish; but, rather, a contrast that even in Catalan is less robust and more variable than other contrasts in the language (see Coveney, 2001; Dufour, Nguyen, & Frauenfelder, 2007; Fagyal, Hassa, & Ngom, 2002; Fagyal, Jenkins, & Kibbee, 2006 for similar findings on the /e/–/ɛ/ contrast in varieties of French).

The perception and production experiments in this study show that even though the Catalan mid-vowel contrasts are less robust than contrasts between other pairs of Catalan vowels, Spanish–Catalan bilinguals in Majorca maintain two independent phonetic categories in the Catalan front and back mid-vowel space, and as a result produce and perceive both Catalan-specific categories

despite the overlap with one phonetic category in Spanish. The differences in the production and perception patterns found in this bilingual community in comparison to previous studies in Barcelona can be attributed to the fact that the Catalan mid-vowel contrasts are acoustically lower and more robust in Majorca (Carrera-Sabaté & Fernández-Planas, 2005; Recasens & Espinosa, 2006, 2009), and that bilinguals in this community may have less exposure to exemplars containing merged mid-vowels. These results suggest that the Catalan vowel categories in the input that Spanish-dominant speakers receive in Majorca are more acoustically distinctive (i.e. more prototypical and with a larger acoustic distance between the mid-vowel categories), ultimately facilitating acquisition. These explanations point to the need for more studies that assess the differences in the acoustic input that bilinguals receive not only as children, but also in their everyday lives as adults, and how it affects their production, perception, and lexical representations.

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Notes

1. For instance the verb "to drink" in Eastern Catalan is pronounced /'bɛwrə/ while the same word is /'bewrə/ in Western Catalan, and /'bəwrə/ in Majorcan Catalan.
2. The participants were not residents of towns that are known to differ from the rest of Majorcan Catalan speakers in their production of Catalan mid-vowels (e.g. Lloseta, Binissalem, Alaró, Sineu, Felanitx, Maria de la Salut, and Sant Joan). In these towns, /ə/ in stressed position has shifted to /ɛ/, as in Barcelona (Moll, 1968). For example, the word pera "pear" is pronounced with the central vowel /'pərə/ in the rest of the island, but /'pɛrə/ in these towns.
3. For more information on the BLP, see Gertken, Amengual, and Birdsong (2014).
4. Taking the median rather than the mean has been claimed to reduce the effect of formant measurement errors (Escudero, Boersma, Rauber, & Bion, 2009).

References

Adank, P., Smits, R., & van Hout, R. (2004). A comparison of vowel normalization procedures for language variation research. *Journal of the Acoustical Society of America*, *116*(5), 3099–3107.

Amengual, M. (2011). Spanish and Catalan in Majorca: Are there contact-induced changes in the Majorcan Catalan vowel system? In L. Ortiz (Ed.), *Selected proceedings of the 13th Hispanic linguistics symposium* (pp. 214–223). Somerville, MA: Cascadilla Proceedings Project.

Amengual, M. (2013). *An experimental approach to phonetic transfer in the production and perception of early Spanish-Catalan bilinguals*. Unpublished doctoral dissertation, University of Texas at Austin.

Birdsong, D., Gertken, L. M., & Amengual, M. (2012). Bilingual language profile: An easy-to-use instrument to assess bilingualism. COERLL, University of Texas at Austin.

Boersma, P., & Weenink, D. (2014). Praat: Doing phonetics by computer (5.3.18). Retrieved from: www.praat.org.

Bosch, L., Costa, A., & Sebastián-Gallés, N. (2000). First and second language vowel perception in early bilinguals. *European Journal of Cognitive Psychology*, *12*(2), 189–221.

Bosch, L., & Ramón-Casas, M. (2011). Variability in vowel production by bilingual speakers: Can input properties hinder the early stabilization of contrastive categories? *Journal of Phonetics*, *39*, 514–526.

Bullock, B. E. (2009). Prosody in contact in French: A case study from a heritage variety in the USA. *International Journal of Bilingualism*, *13*(2), 165–194.

Carrera-Sabaté, J., & Fernández-Planas, A. M. (2005). Vocals mitjanes tòniques del català. *Estudi contrastiu interdialectal*. Barcelona, Spain: Horsori.

Cedrus Corporation (2012). SuperLab (Version 4.5) [Computer software]. San Pedro, CA.

Clopper, C. G. (2009). Computational methods for normalizing acoustic vowel data for talker differences. *Language and Linguistic Compass*, *3*(6), 1430–1442.

Cortés, S., Lleó, C., & Benet, A. (2009). Gradient merging of vowels in Barcelona Catalan under the influence of Spanish. In K. Braunmuller, & J. House (Eds.), *Convergence and divergence in language contact situations* (pp. 185–204). Amsterdam, The Netherlands: John Benjamins.

Coveney, A. (2001). *The sounds of contemporary French: Articulation and diversity*. Exeter, UK: Elm Bank Publications.

DeKeyser, R. (2000). The robustness of critical period effects in second language acquisition. *Studies in Second Language Acquisition*, *22*, 499–534.

Dufour, S., Nguyen, N., & Frauenfelder, U. H. (2007). The perception of phonemic contrasts in a non-native dialect. *Journal of the Acoustical Society of America*, *121*(4), EL131–EL136.

Escudero, P., Boersma, P., Rauber, A., & Bion, R. (2009). A cross-dialectal acoustic description of vowels: Brazilian and European Portuguese. *Journal of the Acoustical Society of America*, *126*, 1379–1393.

Fagyal, Z., Hassa, S., & Ngom, F. (2002). L'opposition [e]-[ɛ] en syllables ouvertes en fin de mot en français parisien: étude acoustique préliminaire. In *Proceedings of the XXIVémes journèes d'études sur la parole* (pp. 165–168). Nancy, France.

Fagyal, Z., Jenkins, F., & Kibbee, D. (2006). *French: A linguistic introduction*. Cambridge, UK: Cambridge University Press.

Flege, J. E. (1991). Age of learning affects the authenticity of voice onset time (VOT) in stop consonants produced in a second language. *Journal of the Acoustic Society of America*, *89*, 395–441.

Flege, J. E. (1993). Production and perception of a novel, second-language phonetic contrast. *Journal of the Acoustical Society of America*, *93*(3), 1589–1608.

Flege, J. E. (1999). Age of learning and second language speech. In D. Birdsong (Ed.), *Second language acquisition and the critical period hypothesis* (pp. 101–132). Hillsdale, NJ: Lawrence Earlbaum.

Flege, J. E. (2003). Assessing constraints on second-language phonetics and phonology. In A. Meyer, & N. Schiller (Eds.), *Phonetics and phonology in language comprehension and production, differences and similarities* (pp. 319–355). Berlin, Germany: Mouton de Gruyter.

Flege, J. E. (2007). Language contact in bilingualism: Phonetic system interactions. In J. Cole, & J. Hualde (Eds.), *Laboratory Phonology 9* (pp. 353–382). Berlin, Germany: Mouton de Gruyter.

Flege, J. E., Bohn, O. S., & Jang, S. (1997). Effects of experience on non-native speakers' production and perception of English vowels. *Journal of Phonetics*, *25*, 437–470.

Flege, J. E., & MacKay, I. R. A. (2004). Perceiving vowels in a second language. *Studies in Second Language Acquisition*, *26*(1), 1–34.

Flege, J. E., MacKay, I. R. A., & Meador, D. (1999). Native Italian speakers' perception and production of English vowels. *Journal of the Acoustical Society of America*, *106*, 2973–2987.

Flege, J. E., Munro, M. J., & MacKay, I. R. A. (1995). Factors affecting strength of perceived foreign accent in a second language. *Journal of the Acoustical Society of America*, *97*(5), 3125–3134.

Fowler, C. A., & Galantucci, B. (2005). The relation of speech perception and speech production. In D. B. Pisoni, & R. E. Remez (Eds.), *The handbook of speech perception* (pp. 633–652). New York, NY: Basil Blackwell.

Gertken, L. M., Amengual, M., & Birdsong, D. (2014). Assessing language dominance with the bilingual language profile. In P. Leclercq, A. Edmonds, & H. Hilton (Eds.), *Measuring L2 proficiency: Perspectives from SLA* (pp. 208–225). Bristol, UK: Multilingual Matters.

Herrick, D. (2003). *An acoustic analysis of phonological vowel reduction in six varieties of Catalan*. Unpublished doctoral dissertation, UC Santa Cruz.

Herrick, D. (2006). Mid vowels and schwa in Eastern Catalan: Five non-Barcelona dialects. In J.- P. Y. Montreuil (Ed.), *New perspectives on Romance linguistics* (pp. 113–126). Amsterdam, The Netherlands: John Benjamins.

Herrick, D. (2007). Eastern Catalan vowel reduction is characterized by raising – not centralization. *Studies in Language Sciences*, *6*, 227–241.

Jenkins, J. J., Strange, W., & Polka, L. (1995). Not everyone can tell a "rock" from a "lock": Assessing individual differences in speech perception. In D. J. Lubinski, & R. V. Dawis (Eds.), *Assessing individual differences in human behavior: New concepts, methods, and findings* (pp. 297–325). Palo Alto, CA: Davies-Black Publishing.

Lenneberg, E. (1967). *Biological foundations of language*. New York, NY: Wiley.

Lleó., C., Benet, A., & Cortés, S. (2007). Some current phonological features in the Catalan of Barcelona. *Catalan Review*, 21, 279–300.

Lleó, C., Benet, A., & Cortés, S. (2009). Límits de la normalització lingüística: vocals vulnerables en el català de Barcelona. In J. Kabatek, & C. D. Pusch (Eds.), *Variation, Polyglossie und Standardisierung* (pp.157–180). Aachen, Germany: Shaker.

Lleó, C., Cortés, S., & Benet, A. (2008). Contact-induced phonological changes in the Catalan spoken in Barcelona. In P. Siemund, & N. Kintana (Eds.), *Language Contact and Contact Languages* (pp.185–212). Amsterdam, The Netherlands: John Benjamins.

Lleó, C., & Rakow, M. (2005). Markedness effects in voiced stop spirantization in bilingual German-Spanish children. In J. Cohen, K. T. McAlister, K. Rolstad, & J. MacSwan (Eds.), *Proceedings of the 4th International Symposium on Bilingualism* (pp.1353–1371). Somerville, MA: Cascadilla Press.

Long, M. (1990). Maturational constraints on language development. *Studies in Second Language Acquisition 12*, 251–285.

MacLeod, A. N., & Stoel-Gammon, C. (2010). What is the impact of age of second language acquisition on the production of consonants and vowels among childhood bilinguals? *International Journal of Bilingualism*, *14*(4), 400–421.

Maddieson, I. (1984). *Patterns of sounds*. Cambridge, UK: Cambridge University Press.

Moll, F. de B. (1968). *Gramàtica catalana referida especialment a les Illes Balears*. Palma de Mallorca, Spain: Moll.

Mora, J. C., Keidel, J. L., & Flege, J. E. (2011). Why are the Catalan contrasts between /e/-/ɛ/ and /o/-/ɔ/ so difficult for even early Spanish-Catalan bilinguals to perceive? In M. Wrembel, M. Kul, & K. Dziubalska-Kołaczyk (Eds.), *Achievements and perspectives in the acquisition of second language speech: New Sounds 2010*, Vol.2 (pp. 183–193). Bern, Switzerland: Peter Lang.

Mora, J. C., & Nadeu, M. (2012). L2 effects on the perception and production of a native vowel contrast in early bilinguals. *International Journal of Bilingualism*, *16*(4), 484–499.

Oyama, S. (1976). A sensitive period for the acquisition of a nonnative phonological system. *Journal of Psycholinguistic Research*, *5*, 261–283.

Pallier, C., Bosch, L., & Sebastián-Gallés, N. (1997). A limit on behavioural plasticity in speech perception. *Cognition*, *64*, B9–B17.

Patkowski, M. (1990). Age and accent in a second language: A reply to James Emil Flege. *Applied Psycholinguistics*, *11*, 73–89.

Piske, T., MacKay, I. R. A., & Flege, J. E. (2001). Factors affecting degree of foreign accent in an L2: A review. *Journal of Phonetics*, *29*, 191–215.

Pisoni, D. (1995). Some thoughts on "normalization" in speech perception. *Research on Spoken Language Processing, Progress Report 20*, 3–30.

Polka, L. (1991). Cross-language speech perception in adults: Phonemic, phonetic and acoustic contributions. *Journal of the Acoustical Society of America*, *89*(6), 2961–2977.

Polka, L. (1992). Characterizing the influence of native experience on adult speech perception. *Perception and Psychophysics*, *52*, 37–52.

Prieto, P. (2004). *Fonètica i fonología. Els sons del català*. Barcelona, Spain: Editorial UOC.

Ramón-Casas, M., Swingley, D., & Sebastián-Gallés, N. (2009). Vowel categorization during word recognition in bilingual toddlers. *Cognitive Psychology*, *59*, 96–121.

Recasens, D. (1991). *Fonètica descriptiva del català*. Barcelona, Spain: Institut d'Estudis Catalans.

Recasens, D. (1993). *Fonètica i fonologia*. Barcelona, Spain: Enciclopèdia Catalana.

Recasens, D., & Espinosa, A. (2006). Dispersion and variability in Catalan vowels. *Speech Communication 48*, 645–666.

Recasens, D., & Espinosa, A. (2009). Dispersion and variability in Catalan five and six peripheral vowel systems. *Speech Communication*, *51*, 240–258.

Sankoff, G. (2013). Linguistic outcomes of bilingualism. In J.K. Chambers, & N. Schilling-Estes (Eds.), *The handbook of language variation and change*. (pp. 501–518). Oxford, UK: Wiley-Blackwell.

Scovel, T. (1988). *A time to speak: A psycholinguistic inquiry into the critical period for human speech*. Cambridge, MA: Newbury House.

Sebastián-Gallés, N., & Díaz, B. (2012). First and second language speech perception: Graded learning. *Language Learning*, *62*, 131–147.

Sebastián-Gallés, N., Echeverría, S., & Bosch, L. (2005). The influence of initial exposure on lexical representation: Comparing early and simultaneous bilinguals. *Journal of Memory and Language*, *52*, 240–255.

Sebastián-Gallés, N., & Soto-Faraco, S. (1999). Online processing of native and non-native phonemic contrasts in early bilinguals. *Cognition*, *72*, 111–123.

Sheldon, A., & Strange, W. (1982). The acquisition of /r/ and /l/ by Japanese learners of English: Evidence that speech production can precede speech perception. *Applied Psycholinguistics*, *3*, 243–264.

Simonet, M. (2011). Production of a Catalan-specific vowel contrast by early Spanish-Catalan bilinguals. *Phonetica*, *68*, 88–110.

Simonet, M. (2014). Phonetic consequences of dynamic cross-linguistic interference in proficient bilinguals. *Journal of Phonetics*, *43*, 26–37.

Snodgrass, J., & Vanderwart, M. (1980). A standardized set of 260 pictures: Norms for name agreement, image agreement, familiarity, and visual complexity. *Journal of Experimental Psychology: Human Learning & Memory*, *6*, 174–215.

Traunmüller, H. (1990). Analytical expressions for the tonotopic sensory scale. *Journal of the Acoustical Society of America*, *88*, 97–100.

Watt, D. J. L., Fabricius, A. H., & Kendall, T. (2010). More on vowels: Plotting and normalizing. In M. Di Paolo, & M. Yaeger-Dror (Eds.), *Sociophonetics: A student's guide* (pp. 107–118). London, UK: Routledge.

Yeni-Komshian, G., Flege, J. E., & Liu, S. (2000). Pronunciation proficiency in the first and second languages of Korean-English bilinguals. *Bilingualism: Language and Cognition*, *3*, 131–150.

Zwicker, E. (1961). Subdivision of the audible frequency range into critical bands. *Journal of the Acoustical Society of America*, *33*(2), 248.

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Appendix 1. Median formant values and Euclidean distance (Bark).

	BLP	*F*1 /e/	*F*2 /e/	*F*1 /ɛ/	*F*2 /ɛ/	*F*1 /o/	*F*2 /o/	*F*1 /ɔ/	*F*2 /ɔ/	ED /e/–/ɛ/	ED /o/–/ɔ/
1	−49.68	6.08	14.01	7.32	12.96	6.19	9.40	7.32	9.91	1.62	1.24
2	−38.42	5.40	12.00	6.65	11.25	6.25	8.31	7.08	9.21	1.46	1.22
3	−99.08	5.33	13.47	6.22	12.53	6.10	9.30	6.36	9.89	1.29	0.63
4	104.71	6.17	13.26	7.37	12.48	6.56	8.99	7.46	9.89	1.43	1.27
5	−114.87	6.09	13.70	6.94	13.33	6.82	9.86	7.57	10.22	1.33	0.67
6	−103.71	6.05	13.84	6.34	13.02	6.84	9.33	7.14	10.44	1.12	0.66
7	−15.44	5.61	14.22	7.10	13.12	5.79	9.55	7.09	10.51	1.85	1.61
8	50.68	5.14	12.57	6.16	11.68	5.21	8.86	6.31	9.91	1.34	1.51
9	12.61	5.24	12.35	6.24	11.64	5.42	8.81	6.20	9.74	1.23	1.20
10	15.26	5.67	14.16	7.01	13.23	6.00	9.39	7.07	10.15	1.63	1.31
11	76.83	5.22	11.88	6.17	11.21	5.59	9.04	6.19	9.19	1.16	0.61
12	−18.07	5.38	12.60	6.70	11.67	5.70	7.95	6.97	9.56	1.61	2.04
13	−2.36	5.96	13.64	6.85	13.03	6.10	9.52	7.34	10.37	1.07	1.49
14	−72.47	5.70	13.55	6.47	12.95	6.45	9.79	6.89	10.00	0.97	0.49
15	−56.85	6.31	13.69	7.86	12.69	6.62	9.17	8.01	10.25	1.83	1.74
16	115.79	6.03	13.84	7.39	12.86	6.04	9.51	7.54	10.63	1.67	1.87
17	37.33	5.38	13.76	7.13	12.75	5.72	9.18	7.26	10.16	2.02	1.82
18	−20.61	5.14	12.03	6.07	11.68	5.37	8.94	6.24	9.51	0.99	1.04
19	61.47	5.46	12.37	6.65	11.56	6.41	8.85	6.50	9.39	1.43	0.54
20	−68.02	6.02	13.50	7.21	12.77	6.21	9.17	7.18	10.35	1.39	1.53
21	133.58	5.38	13.84	7.31	13.15	5.45	8.81	6.96	9.74	2.05	1.77
22	59.58	4.59	13.49	6.06	11.61	4.97	7.89	6.12	9.13	2.38	1.69
23	−23.61	4.83	12.76	5.88	11.83	5.18	8.63	6.35	9.45	1.39	1.42
24	−9.99	5.32	14.24	7.43	12.86	6.22	9.14	7.88	10.36	2.51	2.05
25	−103.71	6.35	13.80	6.49	13.75	6.53	9.11	6.60	9.88	0.14	0.77
26	−41.14	5.20	13.10	6.48	12.18	5.47	9.15	6.57	10.29	1.57	1.58
27	104.16	5.73	13.94	7.10	12.85	5.91	9.19	7.55	10.39	1.75	2.03
28	127.86	5.11	12.02	6.40	11.39	6.01	7.51	6.99	8.71	1.43	1.55
29	−41.60	5.32	14.31	7.06	13.30	5.40	9.87	7.10	10.90	2.01	1.98
30	−63.84	6.18	13.47	6.72	13.20	6.56	9.86	6.88	10.14	0.60	0.41
31	91.81	5.37	12.32	6.24	11.75	5.68	9.19	6.36	9.78	1.04	0.90
32	77.65	4.89	12.88	6.42	12.15	6.24	8.70	7.25	9.61	1.68	1.35
33	−93.44	6.02	12.56	6.30	12.29	6.23	9.05	6.34	9.66	0.38	0.62
34	−68.02	5.95	14.22	7.21	13.47	5.90	9.16	7.20	9.83	1.47	1.46
35	90.54	5.94	14.28	7.64	13.31	6.24	9.20	7.67	10.10	1.96	1.68
36	−58.75	5.36	13.08	5.69	12.54	6.09	8.57	6.04	8.99	0.63	0.42
37	−15.62	5.95	14.63	7.61	11.70	6.48	9.78	7.83	10.44	2.36	1.49
38	−73.47	6.53	12.78	6.67	12.44	6.74	9.96	6.88	10.15	0.36	0.23
39	103.25	5.74	13.83	7.35	12.63	6.14	9.41	7.30	10.25	2.00	1.43
40	66.28	5.27	11.93	6.25	11.15	5.28	8.25	6.20	8.99	1.25	1.18
41	−66.39	5.87	11.40	7.28	11.34	6.37	9.66	7.44	10.26	1.40	1.22
42	107.34	4.80	13.10	6.83	11.74	5.44	8.09	6.60	9.02	2.44	1.48
43	−6.99	5.69	14.11	7.41	13.21	5.81	8.56	7.38	9.73	1.94	1.95
44	13.89	5.53	13.40	6.85	12.64	5.61	9.59	6.81	10.61	1.52	1.56
45	92.54	4.95	13.59	6.23	12.29	5.34	8.58	6.28	9.37	1.82	1.22

Appendix 1. (Continued)

	BLP	*F*1 /e/	*F*2 /e/	*F*1 /ɛ/	*F*2 /ɛ/	*F*1 /o/	*F*2 /o/	*F*1 /ɔ/	*F*2 /ɔ/	ED /e/–/ɛ/	ED /o/–/ɔ/
46	90.54	5.42	13.07	7.21	12.51	5.66	9.07	6.99	9.90	1.86	1.57
47	−51.31	6.29	13.65	6.87	13.16	7.04	9.67	7.13	10.10	0.76	0.43
48	109.42	5.33	12.54	6.81	11.63	5.61	8.41	6.68	9.02	1.73	1.22
49	89.09	5.27	12.39	6.31	11.67	5.52	8.95	6.39	9.49	1.26	1.02
50	111.79	5.04	13.11	6.53	12.17	5.82	8.39	6.71	9.53	1.75	1.44
51	52.50	5.85	13.54	7.20	12.87	6.10	10.3	7.30	10.99	1.51	1.38
52	59.67	5.48	12.73	6.71	11.93	5.86	8.74	6.63	9.60	1.45	1.16
53	30.70	5.36	12.68	6.31	12.15	5.53	9.47	6.52	9.92	1.08	1.09
54	−117.33	6.48	13.52	6.78	13.23	7.08	9.87	7.03	10.42	0.41	0.55
55	58.66	4.81	13.12	6.08	12.24	5.18	7.91	6.15	9.01	1.54	1.46
56	−97.44	6.15	13.80	7.65	12.89	6.82	9.88	7.69	10.45	1.75	1.03
57	−79.55	5.17	12.20	6.09	11.59	5.50	8.78	6.31	9.24	1.10	0.93
58	72.93	5.78	13.90	7.18	12.90	6.28	9.29	7.78	10.24	1.72	1.77
59	−17.80	4.62	12.61	5.96	12.15	5.16	8.31	6.25	9.25	1.41	1.43
60	95.53	5.94	14.22	7.62	12.98	6.54	9.87	7.99	10.75	2.09	1.69

BLP: Bilingual Language Profile; *F*: Formant; ED: Euclidean distance